

Отчёт по лабораторной работе 6

Адресация IPv4 и IPv6. Двойной стек

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Цели и задачи работы

Изучение принципов распределения и настройки адресного пространства на устройствах сети.

Выполнение лабораторной работы

Топология лабораторного стенда в GNS3

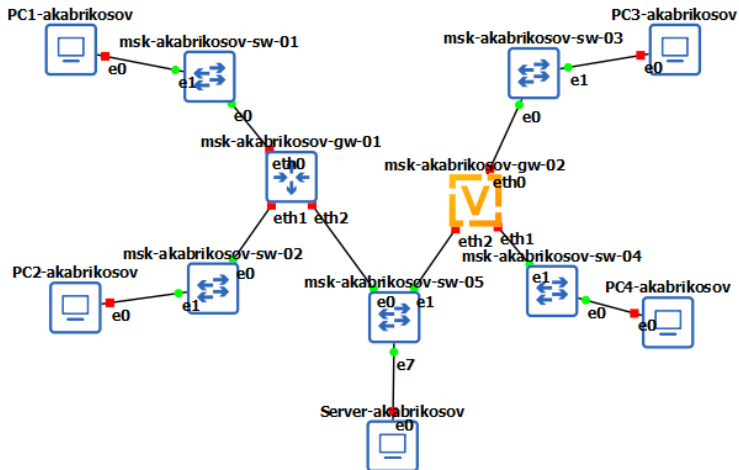


Рис. 1: Топология лабораторного стенда в GNS3

Настройка IPv4-адресации на PC1-akabrikosov

```
PC1-akabrikosov - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 172.16.20.10/25 172.16.20.1
Checking for duplicate address...
VPCS : 172.16.20.10 255.255.255.128 gateway 172.16.20.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 172.16.20.10/25
GATEWAY     : 172.16.20.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 10022
RHOST:PORT  : 127.0.0.1:10023
MTU         : 1500

VPCS>
```

Настройка IPv4-адресации на PC2-akabrikosov

```
PC2-akabrikosov - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 172.16.20.138/25 172.16.20.129
Checking for duplicate address...
VPCS : 172.16.20.138 255.255.255.128 gateway 172.16.20.129

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 172.16.20.138/25
GATEWAY     : 172.16.20.129
DNS         :
MAC         : 00:50:79:66:68:01
LPORT      : 10024
RHOST:PORT  : 127.0.0.1:10025
MTU         : 1500

VPCS>
```

Настройка IPv4-адресации на Server-akabrikosov

```
Server-akabrikosov - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 64.100.1.10/24 64.100.1.1
Checking for duplicate address...
VPCS : 64.100.1.10 255.255.255.0 gateway 64.100.1.1

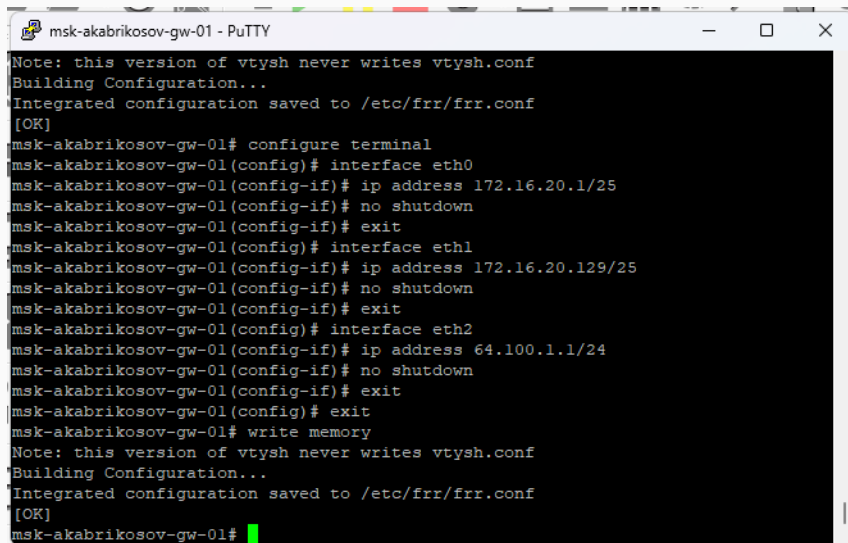
VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 64.100.1.10/24
GATEWAY     : 64.100.1.1
DNS         :
MAC         : 00:50:79:66:68:04
LPORT      : 10026
RHOST:PORT  : 127.0.0.1:10027
MTU         : 1500

VPCS>
```

Настройка интерфейсов маршрутизатора msk-akabrikosov-gw-01



```
msk-akabrikosov-gw-01 - PuTTY
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-akabrikosov-gw-01# configure terminal
msk-akabrikosov-gw-01(config)# interface eth0
msk-akabrikosov-gw-01(config-if)# ip address 172.16.20.1/25
msk-akabrikosov-gw-01(config-if)# no shutdown
msk-akabrikosov-gw-01(config-if)# exit
msk-akabrikosov-gw-01(config)# interface eth1
msk-akabrikosov-gw-01(config-if)# ip address 172.16.20.129/25
msk-akabrikosov-gw-01(config-if)# no shutdown
msk-akabrikosov-gw-01(config-if)# exit
msk-akabrikosov-gw-01(config)# interface eth2
msk-akabrikosov-gw-01(config-if)# ip address 64.100.1.1/24
msk-akabrikosov-gw-01(config-if)# no shutdown
msk-akabrikosov-gw-01(config-if)# exit
msk-akabrikosov-gw-01(config)# exit
msk-akabrikosov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-akabrikosov-gw-01#
```

Рис. 5: Настройка интерфейсов маршрутизатора msk-akabrikosov-gw-01

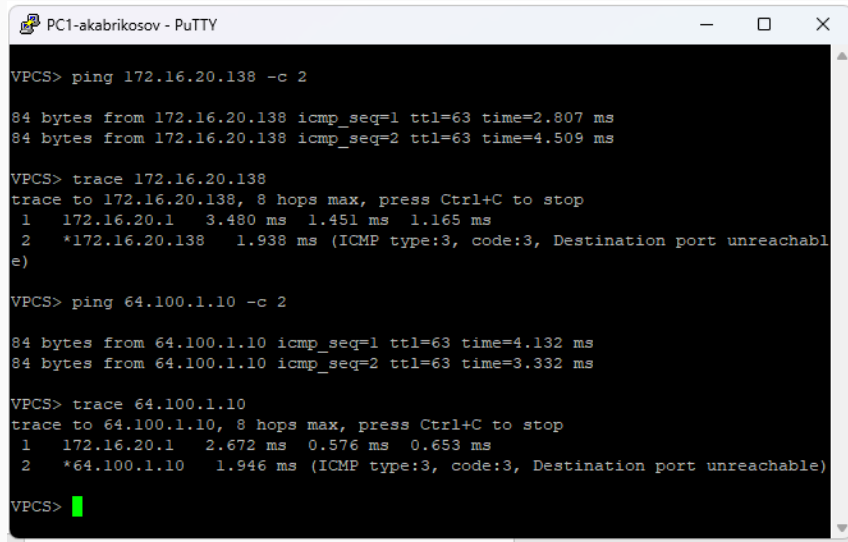
Проверка состояния интерфейсов маршрутизатора

```
msk-akabrikosov-gw-01 - PuTTY

interface eth1
 ip address 172.16.20.129/25
exit
!
interface eth2
 ip address 64.100.1.1/24
exit
!
end
msk-akabrikosov-gw-01# show interface brief
Interface      Status   VRF        Addresses
-----
eth0            up       default    172.16.20.1/25
eth1            up       default    172.16.20.129/25
eth2            up       default    64.100.1.1/24
eth3            down     default
eth4            down     default
eth5            down     default
eth6            down     default
eth7            down     default
lo              up       default
pimreg          up       default

msk-akabrikosov-gw-01#
```

Проверка связности с PC1-akabrikosov



```
PC1-akabrikosov - PuTTY

VPCS> ping 172.16.20.138 -c 2

84 bytes from 172.16.20.138 icmp_seq=1 ttl=63 time=2.807 ms
84 bytes from 172.16.20.138 icmp_seq=2 ttl=63 time=4.509 ms

VPCS> trace 172.16.20.138
trace to 172.16.20.138, 8 hops max, press Ctrl+C to stop
 1  172.16.20.1    3.480 ms  1.451 ms  1.165 ms
 2  *172.16.20.138  1.938 ms (ICMP type:3, code:3, Destination port unreachable)

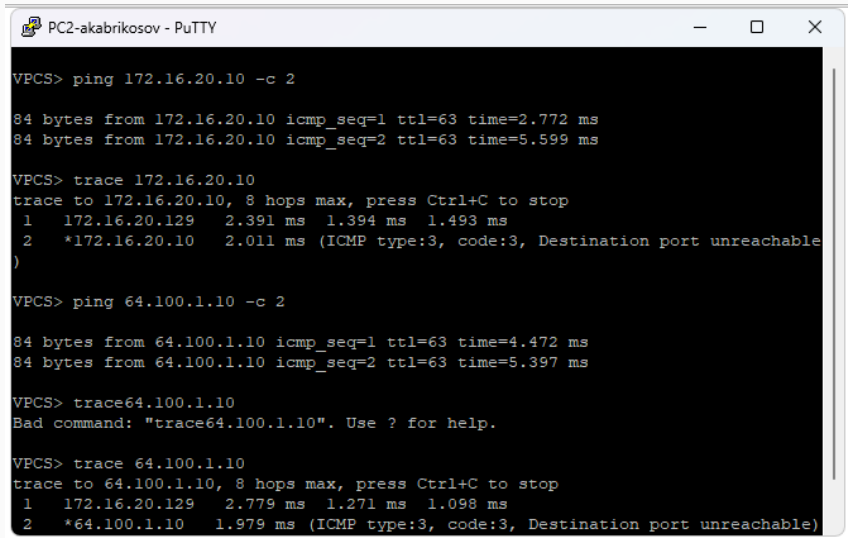
VPCS> ping 64.100.1.10 -c 2

84 bytes from 64.100.1.10 icmp_seq=1 ttl=63 time=4.132 ms
84 bytes from 64.100.1.10 icmp_seq=2 ttl=63 time=3.332 ms

VPCS> trace 64.100.1.10
trace to 64.100.1.10, 8 hops max, press Ctrl+C to stop
 1  172.16.20.1    2.672 ms  0.576 ms  0.653 ms
 2  *64.100.1.10   1.946 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS> █
```

Проверка связности с PC2-akabrikosov



```
PC2-akabrikosov - PuTTY

VPCS> ping 172.16.20.10 -c 2

84 bytes from 172.16.20.10 icmp_seq=1 ttl=63 time=2.772 ms
84 bytes from 172.16.20.10 icmp_seq=2 ttl=63 time=5.599 ms

VPCS> trace 172.16.20.10
trace to 172.16.20.10, 8 hops max, press Ctrl+C to stop
 1  172.16.20.129    2.391 ms  1.394 ms  1.493 ms
 2  *172.16.20.10    2.011 ms (ICMP type:3, code:3, Destination port unreachable)
)

VPCS> ping 64.100.1.10 -c 2

84 bytes from 64.100.1.10 icmp_seq=1 ttl=63 time=4.472 ms
84 bytes from 64.100.1.10 icmp_seq=2 ttl=63 time=5.397 ms

VPCS> trace64.100.1.10
Bad command: "trace64.100.1.10". Use ? for help.

VPCS> trace 64.100.1.10
trace to 64.100.1.10, 8 hops max, press Ctrl+C to stop
 1  172.16.20.129    2.779 ms  1.271 ms  1.098 ms
 2  *64.100.1.10     1.979 ms (ICMP type:3, code:3, Destination port unreachable)
```

Настройка IPv6-адресации на PC3-akabrikosov

```
PC3-akabrikosov - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 2001:db8:c0de:12::a/64
PC1 : 2001:db8:c0de:12::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6802/64
GLOBAL SCOPE        : 2001:db8:c0de:12::a/64
DNS                  :
ROUTER LINK-LAYER   :
MAC                  : 00:50:79:66:68:02
LPORT                : 10044
RHOST:PORT           : 127.0.0.1:10045
MTU                  : 1500

VPCS> █
```

Настройка IPv6-адресации на PC4-akabrikosov

```
PC4-akabrikosov - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 2001:db8:c0de:13::a/64
PC1 : 2001:db8:c0de:13::a/64

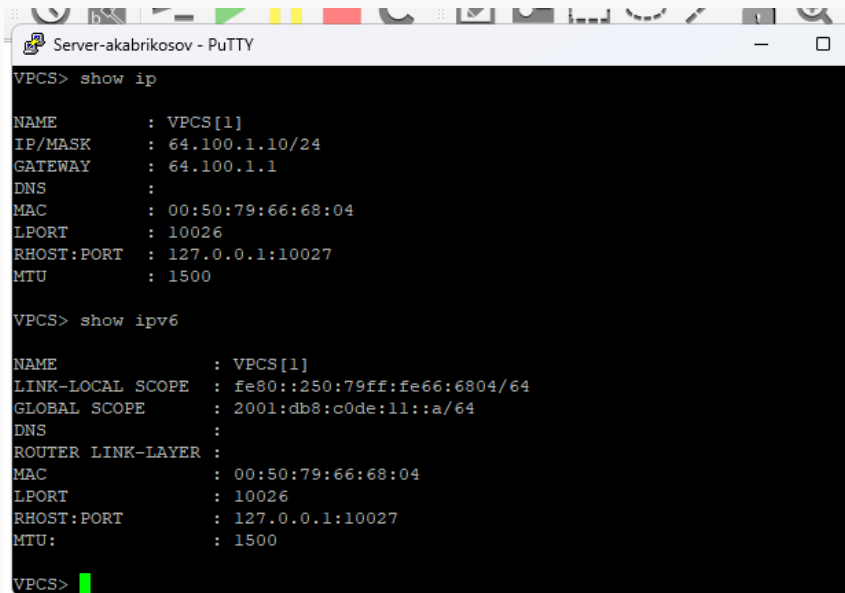
VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6803/64
GLOBAL SCOPE        : 2001:db8:c0de:13::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:03
LPORT               : 10046
RHOST:PORT          : 127.0.0.1:10047
MTU                 : 1500

VPCS> 
```

Проверка IPv4- и IPv6-адресации на Server-akabrikosov



```
Server-akabrikosov - PuTTY
VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 64.100.1.10/24
GATEWAY    : 64.100.1.1
DNS        :
MAC        : 00:50:79:66:68:04
LPORT      : 10026
RHOST:PORT : 127.0.0.1:10027
MTU        : 1500

VPCS> show ipv6

NAME           : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6804/64
GLOBAL SCOPE   : 2001:db8:c0de:11::a/64
DNS            :
ROUTER LINK-LAYER :
MAC            : 00:50:79:66:68:04
LPORT          : 10026
RHOST:PORT     : 127.0.0.1:10027
MTU            : 1500

VPCS> 
```

Настройка IPv6-интерфейсов маршрутизатора VyOS

```
msk-akabrikosov-gw-02 - PuTTY
[edit]
vyos@msk-akabrikosov-gw-02# commit
[edit]
vyos@msk-akabrikosov-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-akabrikosov-gw-02# show interfaces
  ethernet eth0 {
    address 2001:db8:c0de:12::1/64
    hw-id 0c:21:67:b7:00:00
  }
  ethernet eth1 {
    address 2001:db8:c0de:13::1/64
    hw-id 0c:21:67:b7:00:01
  }
  ethernet eth2 {
    address 2001:db8:c0de:11::1/64
    hw-id 0c:21:67:b7:00:02
  }
  loopback lo {
  }
[edit]
vyos@msk-akabrikosov-gw-02#
```

Проверка интерфейсов маршрутизатора VyOS

```
PC3-akabrikosov - PuTTY

VPCS> ping 2001:db8:c0de:11::a -c 2

2001:db8:c0de:11::a icmp6_seq=1 ttl=62 time=5.975 ms
2001:db8:c0de:11::a icmp6_seq=2 ttl=62 time=2.349 ms

VPCS> ping 2001:db8:c0de:13::a -c 2

2001:db8:c0de:13::a icmp6_seq=1 ttl=62 time=3.695 ms
2001:db8:c0de:13::a icmp6_seq=2 ttl=62 time=2.497 ms

VPCS> trace 2001:db8:c0de:11::a

trace to 2001:db8:c0de:11::a, 64 hops max
 1 2001:db8:c0de:12::1    2.534 ms   1.477 ms   1.744 ms
 2 2001:db8:c0de:11::a    2.384 ms   1.681 ms   1.370 ms

VPCS> trace 2001:db8:c0de:13::a

trace to 2001:db8:c0de:13::a, 64 hops max
 1 2001:db8:c0de:12::1    2.787 ms   2.015 ms   0.985 ms
 2 2001:db8:c0de:13::a    3.239 ms   2.401 ms   1.276 ms

VPCS>
```

Проверка IPv6-связности с PC3-akabrikosov

```
PC4-akabrikosov - PuTTY

VPCS> ping 2001:db8:c0de:12::a -c 2

2001:db8:c0de:12::a icmp6_seq=1 ttl=62 time=4.994 ms
2001:db8:c0de:12::a icmp6_seq=2 ttl=62 time=2.010 ms

VPCS> ping 2001:db8:c0de:11::a -c 2

2001:db8:c0de:11::a icmp6_seq=1 ttl=62 time=5.166 ms
2001:db8:c0de:11::a icmp6_seq=2 ttl=62 time=2.702 ms

VPCS> trace 2001:db8:c0de:12::a

a
trace to 2001:db8:c0de:12::a, 64 hops max
 1 2001:db8:c0de:13::1    2.097 ms   0.754 ms   0.959 ms
 2 2001:db8:c0de:12::a    1.567 ms   0.997 ms   0.766 ms

VPCS> trace 2001:db8:c0de:11::a

trace to 2001:db8:c0de:11::a, 64 hops max
 1 2001:db8:c0de:13::1    3.410 ms   0.918 ms   0.912 ms
 2 2001:db8:c0de:11::a    2.833 ms   1.191 ms   1.308 ms

VPCS> █
```

Проверка IPv6-связности с PC4-akabrikosov

```
PC4-akabrikosov - PuTTY
trace to 2001:db8:c0de:12::a, 64 hops max
 1 2001:db8:c0de:13::1    2.097 ms    0.754 ms    0.959 ms
 2 2001:db8:c0de:12::a    1.567 ms    0.997 ms    0.766 ms

VPCS> trace 2001:db8:c0de:11::a

trace to 2001:db8:c0de:11::a, 64 hops max
 1 2001:db8:c0de:13::1    3.410 ms    0.918 ms    0.912 ms
 2 2001:db8:c0de:11::a    2.833 ms    1.191 ms    1.308 ms

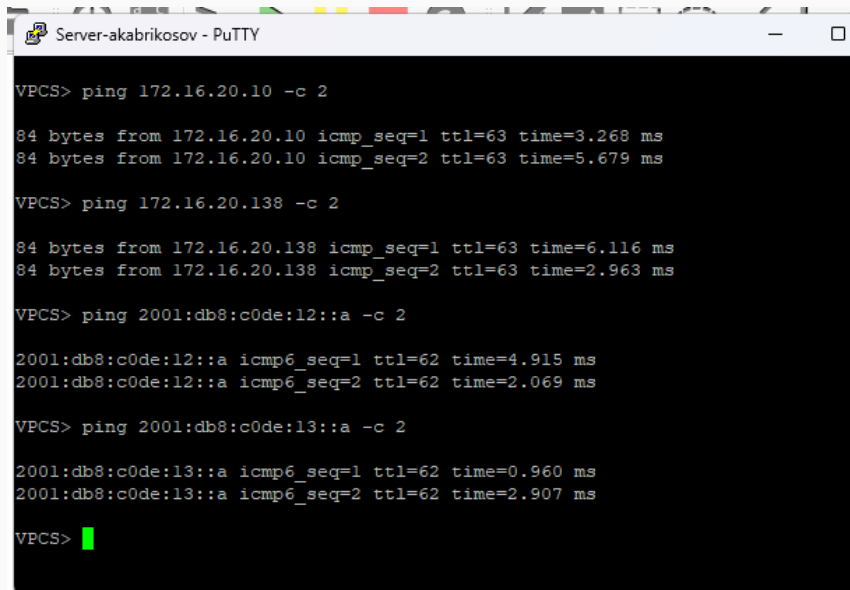
VPCS>
VPCS> ping 172.16.20.10
host (172.16.20.10) not reachable

VPCS> ping 172.16.20.138
host (172.16.20.138) not reachable

VPCS> ping 64.100.1.10
host (64.100.1.10) not reachable

VPCS>
```

Проверка изоляции IPv4 и IPv6 подсетей



```
Server-akabrikosov - PuTTY

VPCS> ping 172.16.20.10 -c 2

84 bytes from 172.16.20.10 icmp_seq=1 ttl=63 time=3.268 ms
84 bytes from 172.16.20.10 icmp_seq=2 ttl=63 time=5.679 ms

VPCS> ping 172.16.20.138 -c 2

84 bytes from 172.16.20.138 icmp_seq=1 ttl=63 time=6.116 ms
84 bytes from 172.16.20.138 icmp_seq=2 ttl=63 time=2.963 ms

VPCS> ping 2001:db8:c0de:12::a -c 2

2001:db8:c0de:12::a icmp6_seq=1 ttl=62 time=4.915 ms
2001:db8:c0de:12::a icmp6_seq=2 ttl=62 time=2.069 ms

VPCS> ping 2001:db8:c0de:13::a -c 2

2001:db8:c0de:13::a icmp6_seq=1 ttl=62 time=0.960 ms
2001:db8:c0de:13::a icmp6_seq=2 ttl=62 time=2.907 ms

VPCS> 
```

Захват ARP-трафика на интерфейсе сервера

16	65.519689	64.100.1.10	172.16.20.138	ICMP	98 Echo (ping) reply	id=0x4d15, seq=2/512, t
17	69.521552	0c:46:43:8b:00:02	Private_66:68:04	ARP	60 Who has 64.100.1.10? Tell 64.100.1.1	
18	69.523287	Private_66:68:04	0c:46:43:8b:00:02	ARP	60 64.100.1.10 is at 00:50:79:66:68:04	
19	74.966498	172.16.20.138	64.100.1.10	UDP	106 16032 → 16033 Len=64	
20	74.967177	64.100.1.10	172.16.20.138	ICMP	86 Destination unreachable (Port unreachable)	

> Frame 17: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

▼ Ethernet II, Src: 0c:46:43:8b:00:02 (0c:46:43:8b:00:02), Dst: Private_66:68:04 (00:50:79:66:68:04)

- > Destination: Private_66:68:04 (00:50:79:66:68:04)
- > Source: 0c:46:43:8b:00:02 (0c:46:43:8b:00:02)
- Type: ARP (0x0806)
- [Stream index: 1]
- Padding: 00000000000000000000000000000000

▼ Address Resolution Protocol (request)

- Hardware type: Ethernet (1)
- Protocol type: IPv4 (0x0800)
- Hardware size: 6
- Protocol size: 4
- Opcode: request (1)
- Sender MAC address: 0c:46:43:8b:00:02 (0c:46:43:8b:00:02)
- Sender IP address: 64.100.1.1
- Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
- Target IP address: 64.100.1.10

0000 00 5
0010 08 0
0020 00 0
0030 00 0

Рис. 17: Захват ARP-трафика на интерфейсе сервера

Захват ICMP-трафика IPv4

No.	Time	Source	Destination	Protocol	Length	Info
73	810.222231	2001:db8:c0de:11::a	2001:db8:c0de:13::a	ICMPv6	174	Destination Unreachable (Port unreachable)[M
74	887.273170	Private_66:68:04	Broadcast	ARP	64	Who has 64.100.1.1? Tell 64.100.1.10
75	887.275120	0c:46:43:8b:00:02	Private_66:68:04	ARP	60	64.100.1.1 is at 0c:46:43:8b:00:02
→ 76	887.276558	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) request id=0x8318, seq=1/256, t
← 77	887.279612	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) reply id=0x8318, seq=1/256, t
78	888.281806	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) request id=0x8418, seq=2/512, t
79	888.286250	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) reply id=0x8418, seq=2/512, t
80	892.327048	0c:46:43:8b:00:02	Private_66:68:04	ARP	60	Who has 64.100.1.10? Tell 64.100.1.1

> Frame 77: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

▼ Ethernet II, Src: 0c:46:43:8b:00:02 (0c:46:43:8b:00:02), Dst: Private_66:68:04 (00:50:79:66:68:04)

- > Destination: Private_66:68:04 (00:50:79:66:68:04)
- > Source: 0c:46:43:8b:00:02 (0c:46:43:8b:00:02)
- Type: IPv4 (0x0800)

[Stream index: 1]

▼ Internet Protocol Version 4, Src: 172.16.20.10, Dst: 64.100.1.10

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

- > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 84

Identification: 0x1883 (6275)

- > 000. = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 63

Protocol: ICMP (1)

Header Checksum: 0x619e [validation disabled]

[Header checksum status: Unverified]

Source Address: 172.16.20.10

Destination Address: 64.100.1.10

[Stream index: 0]

▼ Internet Control Message Protocol

0000 00 50

0010 00 54

0020 01 0a

0030 0e 0f

0040 1e 1f

0050 2e 2f

0060 3e 3f

Захват ICMPv6-трафика

No.	Time	Source	Destination	Protocol	Length	Info
82	895.346553	64.100.1.10	172.16.20.138	ICMP	98	Echo (ping) request id=0x8b18, seq=1/256, t
83	895.352141	172.16.20.138	64.100.1.10	ICMP	98	Echo (ping) reply id=0x8b18, seq=1/256, t
84	896.353718	64.100.1.10	172.16.20.138	ICMP	98	Echo (ping) request id=0x8c18, seq=2/512, t
85	896.356249	172.16.20.138	64.100.1.10	ICMP	98	Echo (ping) reply id=0x8c18, seq=2/512, t
→ 86	913.776325	2001:db8:c0de:11::a	2001:db8:c0de:12::a	ICMPv6	118	Echo (ping) request id=0x9d18, seq=1, hop li
← 87	913.780540	2001:db8:c0de:12::a	2001:db8:c0de:11::a	ICMPv6	118	Echo (ping) reply id=0x9d18, seq=1, hop limi
88	914.782563	2001:db8:c0de:11::a	2001:db8:c0de:12::a	ICMPv6	118	Echo (ping) request id=0x9d18, seq=2, hop li
89	914.784255	2001:db8:c0de:12::a	2001:db8:c0de:11::a	ICMPv6	118	Echo (ping) reply id=0x9d18, seq=2, hop limi

> Frame 86: Packet, 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface -, id 0

▼ Ethernet II, Src: Private_66:68:04 (00:50:79:66:68:04), Dst: 0c:21:67:b7:00:02 (0c:21:67:b7:00:02)

> Destination: 0c:21:67:b7:00:02 (0c:21:67:b7:00:02)

> Source: Private_66:68:04 (00:50:79:66:68:04)

Type: IPv6 (0x86dd)

[Stream index: 8]

▼ Internet Protocol Version 6, Src: 2001:db8:c0de:11::a, Dst: 2001:db8:c0de:12::a

0110 = Version: 6

> 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)

.... 0000 0000 0000 0000 = Flow Label: 0x00000

Payload Length: 64

Next Header: ICMPv6 (58)

Hop Limit: 64

> Source Address: 2001:db8:c0de:11::a

> Destination Address: 2001:db8:c0de:12::a

[Stream index: 8]

▼ Internet Control Message Protocol v6

Type: Echo (ping) request (128)

Code: 0

Checksum: 0x0df2 [correct]

[Checksum Status: Good]

Identifier: 0x9d18

Sequence: 1

0000 0c 21

0010 00 00

0020 00 00

0030 00 00

0040 02 03

0050 12 13

0060 22 23

0070 32 33

Топология сети с двумя локальными подсетями

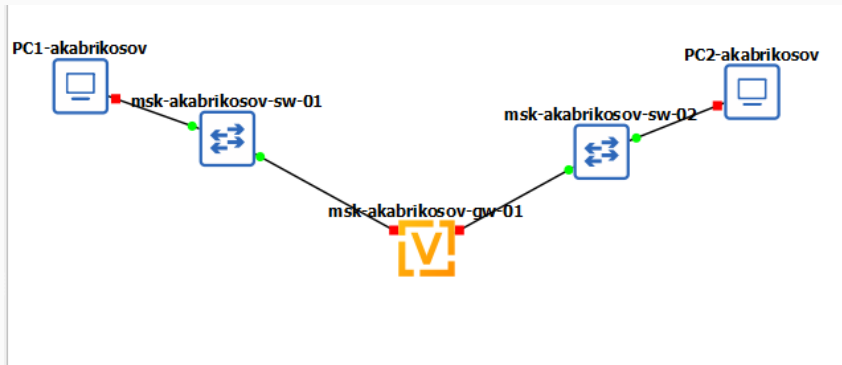


Рис. 20: Топология сети с двумя локальными подсетями

Проверка IPv4 и IPv6 адресации на PC1-akabrikosov

```
PC1-akabrikosov - PuTTY
VPCS> show ip
NAME           : VPCS[1]
IP/MASK        : 10.10.1.100/27
GATEWAY        : 10.10.1.97
DNS            :
MAC            : 00:50:79:66:68:00
LPORT         : 10008
RHOST:PORT     : 127.0.0.1:10009
MTU            : 1500

VPCS> show ipv6
NAME           : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE    : 2001:db8:1:1::a/64
DNS            :
ROUTER LINK-LAYER :
MAC            : 00:50:79:66:68:00
LPORT         : 10008
RHOST:PORT     : 127.0.0.1:10009
MTU            : 1500

VPCS>
```

Проверка IPv4 и IPv6 адресации на PC2-akabrikosov

```
PC2-akabrikosov - PuTTY
VPCS> show ip
NAME      : VPCS[1]
IP/MASK    : 10.10.1.20/28
GATEWAY    : 10.10.1.17
DNS        :
MAC        : 00:50:79:66:68:01
LPORT     : 10010
RHOST:PORT : 127.0.0.1:10011
MTU        : 1500

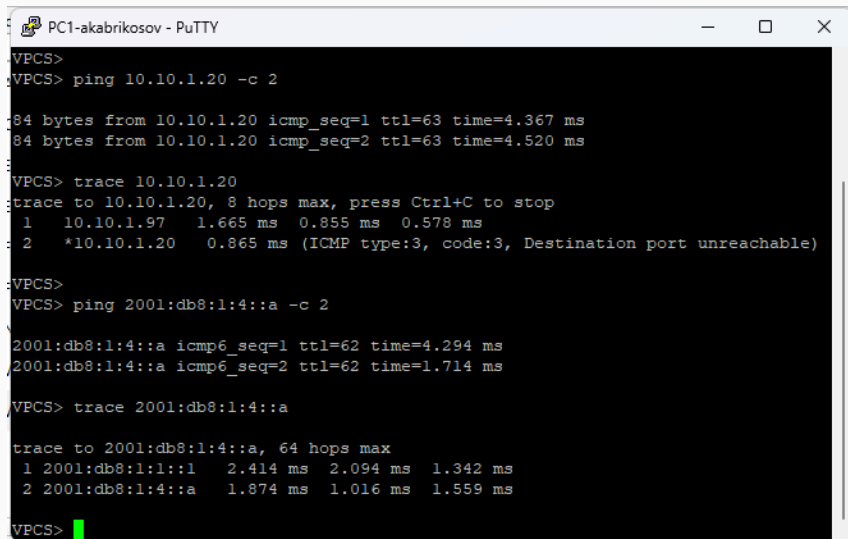
VPCS> show ipv6
NAME      : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE    : 2001:db8:1:4::a/64
DNS            :
ROUTER LINK-LAYER :
MAC           : 00:50:79:66:68:01
LPORT        : 10010
RHOST:PORT    : 127.0.0.1:10011
MTU           : 1500

VPCS>
```

Проверка настроенной адресации интерфейсов VyOS

```
msk-akabrikosov-gw-01 - PuTTY
vyos@msk-akabrikosov-gw-01# commit
[edit]
vyos@msk-akabrikosov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-akabrikosov-gw-01# show interfaces
  ethernet eth0 {
    address 10.10.1.97/27
    address 2001:db8:1:1::1/64
    hw-id 0c:7e:0c:d0:00:00
  }
  ethernet eth1 {
    address 10.10.1.17/28
    address 2001:db8:1:4::1/64
    hw-id 0c:7e:0c:d0:00:01
  }
  ethernet eth2 {
    hw-id 0c:7e:0c:d0:00:02
  }
  loopback lo {
  }
[edit]
vyos@msk-akabrikosov-gw-01#
```

Проверка связности ping и trace между PC1 и PC2



```
PC1-akabrikosov - PuTTY
VPCS>
VPCS> ping 10.10.1.20 -c 2

64 bytes from 10.10.1.20 icmp_seq=1 ttl=63 time=4.367 ms
64 bytes from 10.10.1.20 icmp_seq=2 ttl=63 time=4.520 ms

VPCS> trace 10.10.1.20
trace to 10.10.1.20, 8 hops max, press Ctrl+C to stop
 1  10.10.1.97    1.665 ms  0.855 ms  0.578 ms
 2  *10.10.1.20   0.865 ms (ICMP type:3, code:3, Destination port unreachable)

VPCS>
VPCS> ping 2001:db8:1:4::a -c 2

2001:db8:1:4::a icmp6_seq=1 ttl=62 time=4.294 ms
2001:db8:1:4::a icmp6_seq=2 ttl=62 time=1.714 ms

VPCS> trace 2001:db8:1:4::a

trace to 2001:db8:1:4::a, 64 hops max
 1 2001:db8:1:1::1  2.414 ms  2.094 ms  1.342 ms
 2 2001:db8:1:4::a  1.874 ms  1.016 ms  1.559 ms

VPCS> 
```

Рис. 24: Проверка связности ping и trace между PC1 и PC2

Выводы по проделанной работе

В ходе работы: - выполнено разбиение IPv4 и IPv6 адресных пространств на подсети; - настроен двойной стек IPv4/IPv6 в GNS3; - подтверждена связность внутри соответствующих подсетей; - подтверждена изоляция IPv4 и IPv6 сегментов (кроме Dual Stack сервера); - выполнен анализ ARP, ICMP, ICMPv6 в захваченном трафике.